MINISTRY OF EDUCATION AND VOCATIONAL TRAINING

CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

URBAN WEST REGION - ZANZIBAR

FORM FOUR (IV) MOCK EXAMINATION - 2025

BIOLOGY I

(For Both School and Private Candidates)

033/1

TIME: 3:00 HOURS **JUNE 2025**

INSTRUCTIONS:

1. This paper consists of sections **A, B** and **C** with a total of **eleven (11)** questions.
2. Answer **all** the questions in sections **A,** and **B,** and **two (2)** questions from section **C**.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section **C** carries **thirty (30)** marks.
4. All writing should be in **blue** or **black** pen, **except** for diagrams that must be drawn in pencil.
5. Communication devices and any unauthorized materials are **not allowed** in the examination room.
6. Write your **Examination Number** ON every page of your answer booklet(s).

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SECTION A: (16 Marks)

Answer all questions in this section.

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

(i.) Can be caused by burning plastic waste materials

A. Bruised skin B. Joint pain C. Heart problem
D.Breathing problem E. Headache

(ii.) Receiving signals from the nervous system that prompt to either speed up or slow down the heart rate

A. Myofibril B. Pacemarkers C. Myosis
D. Cardiac muscle E . Action

(iii.) Influencing the formation of buds, cell division enlargement and initiation of root formation in plants

A. Ethylene B. Auxin C. Cytokinin
D. Cardiac muscle E . Action

(iv.) Highly concentration of uric acid in plasma

A. Gout B. Liver cirrhosis C. Diabetes insipidus
D.Urinary tract obstruction E . Uraemia

(v.) It provides all the information the organism requires to function.

A. Trait B. Allele C. Genome D. Gene E . DNA

(vi.) A stage in which microtubules called spindle fibres develop form centrioles

A. Metaphase B. Interphase C. Anaphase D. Prophase E . Telophase

(vii.) Interaction that is beneficial to one organism and is neutral to other organism

A. Mutualism B. Commensalism C. Neutralism
D.Amensalism E . Parasitism

(viii.) Setting the time and rate of cardiac muscle contraction

A. Sinoatrial node B. Tricuspid valve C.Bicuspid valve
D.Semicircular valve E . Atrium

(ix.) It is made up of two layers of lipid and protein molecules

A. Cytoplasm B. Vacuoles C. Cell membrane
D. Cell wall E . Mitochondria

(x.) Example of fat soluble vitamins

A. D B. B12 C. B6 D. C E. K

2. Match the warning signs that are found on apparatus and chemicals container in the laboratory in **List A** with their corresponding components in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided.

S/N	LIST A:	LIST B:
(i.)	Can cause body wound or burn various things such as clothes and furniture's.	**A.** Explosive **B.** Caution
(ii.)	Can cause fire accidents	**C.** Fragile
(iii.)	Can cause death if touched, swallowed or inhaled.	**D.** Toxic
(iv.)	Fire can erupt or can cause health related problems to the smoker and non smoker	**E.** High voltage **F.** Flammable
(v.)	Accident can occurvi(y)Items can easily break and result into a loss	**G.** Corrosive **H.** No smoking

SECTION B: (54 Marks)

Answer all question in this section

3. (a) Why is it necessary to cover your hands when giving first aid to a bleeding person?

   (b) Describe the effects of a waste disposal in the community. (4 effects)

4. Madam Hajra is 25 years old but she looks like a person with 50 years of age. Which factors could have lead to such appearance of Madam Hajra? Provide six (6) points.

5. **A person sit for long time during the class, after finishing the lesson he decide to go home but fail to do so since one among his hind limb contract without relaxing and cause much pain, Apply the knowledge of movement to help this person to walk well and got home.**

6. **(a) Briefly outline four (4) reasons why people become dependent on drugs**

   (b) Why alcohol is a drug?

7. (a) Explain how carbon dioxide is excreted from plants.

   (b) Plants do not need an elaborate excretory system similar to that of animals. Explain.

8. During the study tour in Manyara National Park for form 3 students, they cross with animals such as antelopes, giraffes, lions and leopards climbing the big trees. Also they were able to see grasses and shrub of different size and large group of mushrooms along the lake Manyara, From this tour

(a) (i) Construct a food web

(ii) From the food web, indicate

Primary producers

Primary consumers

Secondary consumers

Decomposer

(b) What will happen if mushroom destroyed from the national park?

SECTION C (30 Marks)

Answer two (2) questions from this section.

9. Among of the diseases causing more death in Tanzania is Malaria although dwellers normally ignore due to the fact that its symptoms may not torture the body instantly, as a biologist invited to educate Tanzania through the national gazette, show the cause, preventive measure of the disease

10. Without blood vertebrate animals can not survive, explain the reasons by giving six points.

11. With the help of a well labelled diagram, Describe the structure of the DNA molecule.